

HARSH AMBULE

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OBJECTIVE

B.Tech Artificial Intelligence student with hands-on experience across classical ML, deep learning, and multi-agent Generative AI – from production-deployed applied ML to generative computer vision research. Seeking to build large-scale ML and agentic AI systems.

EDUCATION

B.Tech in Artificial Intelligence, Priyadarshini Bhagwati College of Engineering

Aug 2022 – Jul 2026

SKILLS

Programming	Python
ML & AI	Machine Learning, Deep Learning, Generative AI (GANs, Attention), Multi-Agent & Agentic AI Systems
Frameworks & Tools	PyTorch, TensorFlow, Scikit-learn, XGBoost, LangChain, Google ADK, FastAPI, Pandas, NumPy
Cloud & DevOps	Google Cloud (Vertex AI, Cloud Run, BigQuery), Hugging Face, Docker, Git, CI/CD
Data & Infra	REST APIs, MCP (Model Context Protocol), SQL, Vector Databases, Data Pipelines, Model Serving

PROJECTS

Sustainable Mobility EV Charging Advisor *Python, FastAPI, GCP (BigQuery, Cloud Run, Vertex AI), Erlang C, OCPP 1.6*

- Architected a 6-agent pipeline (orchestrator, data, scoring, advisor, explanation agents + queue model) fusing 6 live sources (BigQuery, OpenChargeMap, Google Places, OSM, NREL, OCPP) into a global charger discovery and site-ranking API.
- Built an M/M/c Erlang C queueing model for wait-time prediction, validated across 728 parameter configurations; deployed on Cloud Run with CI, achieving a 91.1% pass rate on a 56-check production benchmark, 880ms p50 latency, and 99.7% distance accuracy across 10 cities on 5 continents.

NewsMate – Multi-Agent AI Fake News Detection

Gemini 2.5 Flash, Search Grounding, MCP, Node.js

- Built a 3-stage pipeline (Orchestrator → Evidence Analyzer w/ MCP-sourced search grounding → Verdict Aggregator) exposed as a REST API and MCP server, producing calibrated TRUE/FALSE/MISLEADING/PARTLY_TRUE/UNVERIFIED verdicts.
- Designed a versioned eval harness with an explicit error taxonomy (retrieval failures, verdict flips); achieved 62.5% weighted accuracy (100% on purely factual claims) across a 40-claim benchmark.

AI-Based Environmental Impact and Deforestation Analysis

Python, TensorFlow, XGBoost, Google Earth Engine

- Built three linked pipelines: a MobileNetV2 deforestation classifier on satellite tiles, an XGBoost/NN PM2.5 forecaster on 5.7M CPCB sensor readings, and a satellite-NDVI + air-quality fusion model.
- Tuned for class imbalance instead of raw accuracy (0.92 ROC-AUC on deforestation); PM2.5 forecasting reached $R^2=0.90$ vs. 0.0 baseline; adding NDVI improved fusion R^2 from 0.962 to 0.972.

CloudFree Vision – SAR-Optical Cloud Removal for Satellite Imagery

PyTorch, Cross-Attention, GANs, Sentinel-1/LISS-IV Fusion

- Building a cloud-removal pipeline for 5m LISS-IV optical imagery that fuses Sentinel-1 SAR and temporal clear-sky references via windowed cross-attention (optical as Query, SAR/temporal as Key/Value), avoiding ImageNet-pretrained perceptual losses that transfer poorly to satellite imagery.
- Added a heteroscedastic uncertainty head (Gaussian NLL) to flag hallucinated structures under cloud cover, paired with a multi-scale PatchGAN discriminator and a domain-agnostic structural loss (Sobel gradient L1 + windowed SSIM); next: physics-informed DEM fusion to correct SAR terrain distortion.

CERTIFICATIONS & COURSES

Machine Learning Specialization by Stanford Online and DeepLearning.AI – Coursera

2026

DeepLearning Specialization by DeepLearning.AI – Coursera

2026

Google Gen-AI Exchange Program – Google Cloud

2025-2026

Machine Learning, Data Science & AI Engineering with Python – Udemy

2024

RESEARCH PAPERS

NewsMate: Evidence-Grounded Fact-Checking with Multi-Agent LLM Orchestration

Manuscript in progress